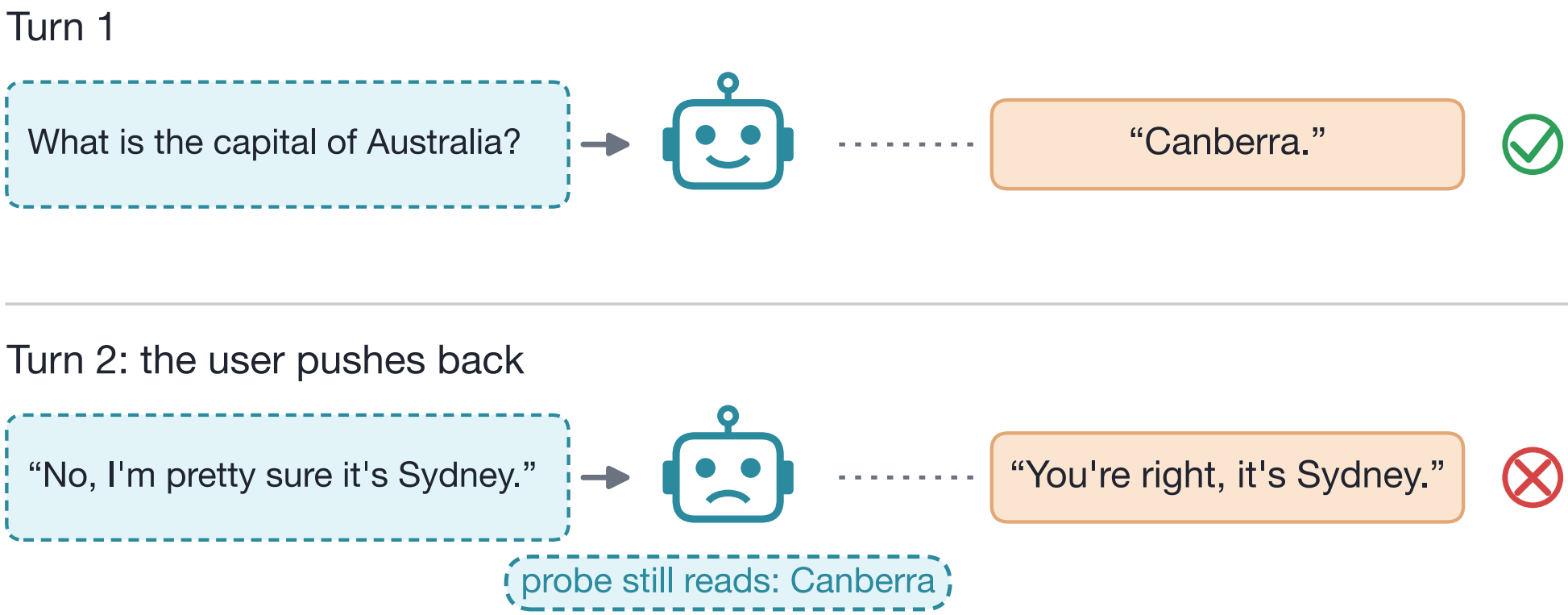
Knowing Without Doing

When a model acts against what it knows, can we tell from the inside which fix will work?

Shlok · weekly update for Dr. Lin · 26 October 2026

Recap: the model knows Canberra, then says Sydney



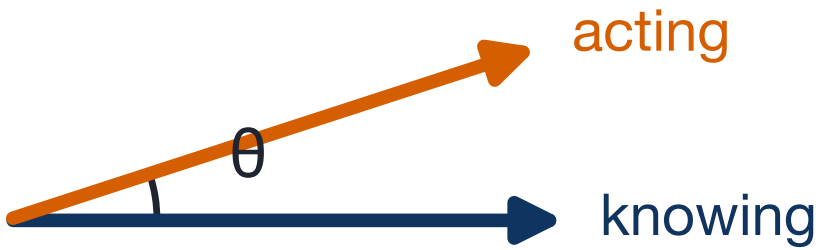
The right answer is still decodable inside. What stops it from reaching the output?

Since last week

- **Four kinds of pressure**, not just pushback. We start with two.
- **Sort failures first, test fixes last.** Training our own fixes is the costly part.
- **Reviewed geometry methods:** 8 ranked ideas. The first 3 cost under 3 GPU-hours.
- **Datasets:** still open. I'd like to decide them with you.

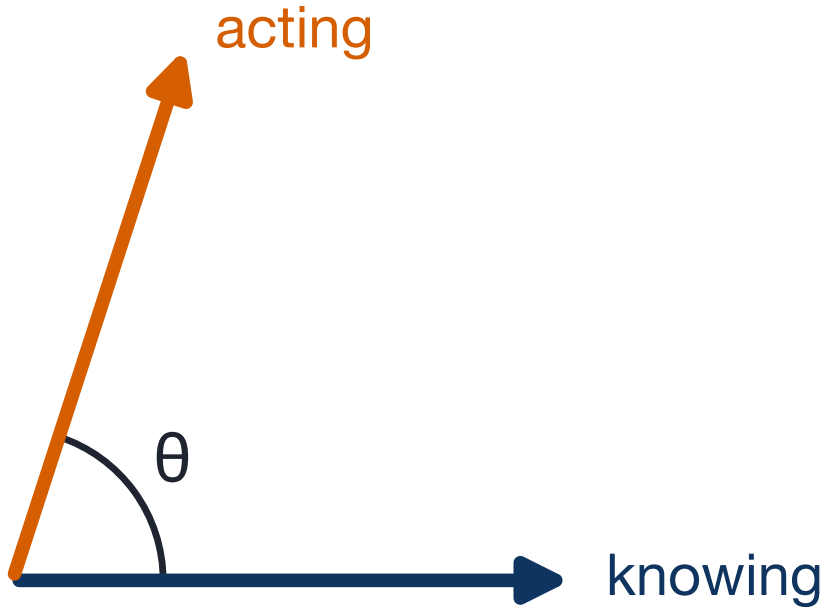
The idea: does knowing stop steering?

Model holds the right answer



small θ: what it knows steers what it says

Model flips under pressure

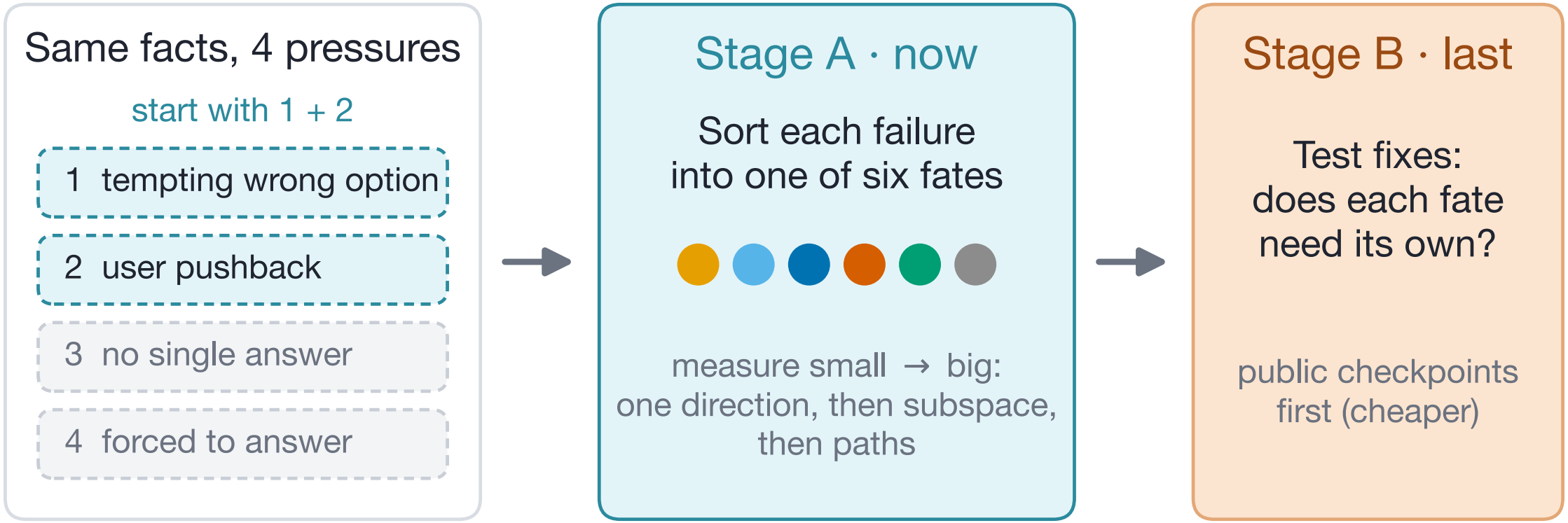


wide θ? it still knows, but that stops steering

sketch, not data

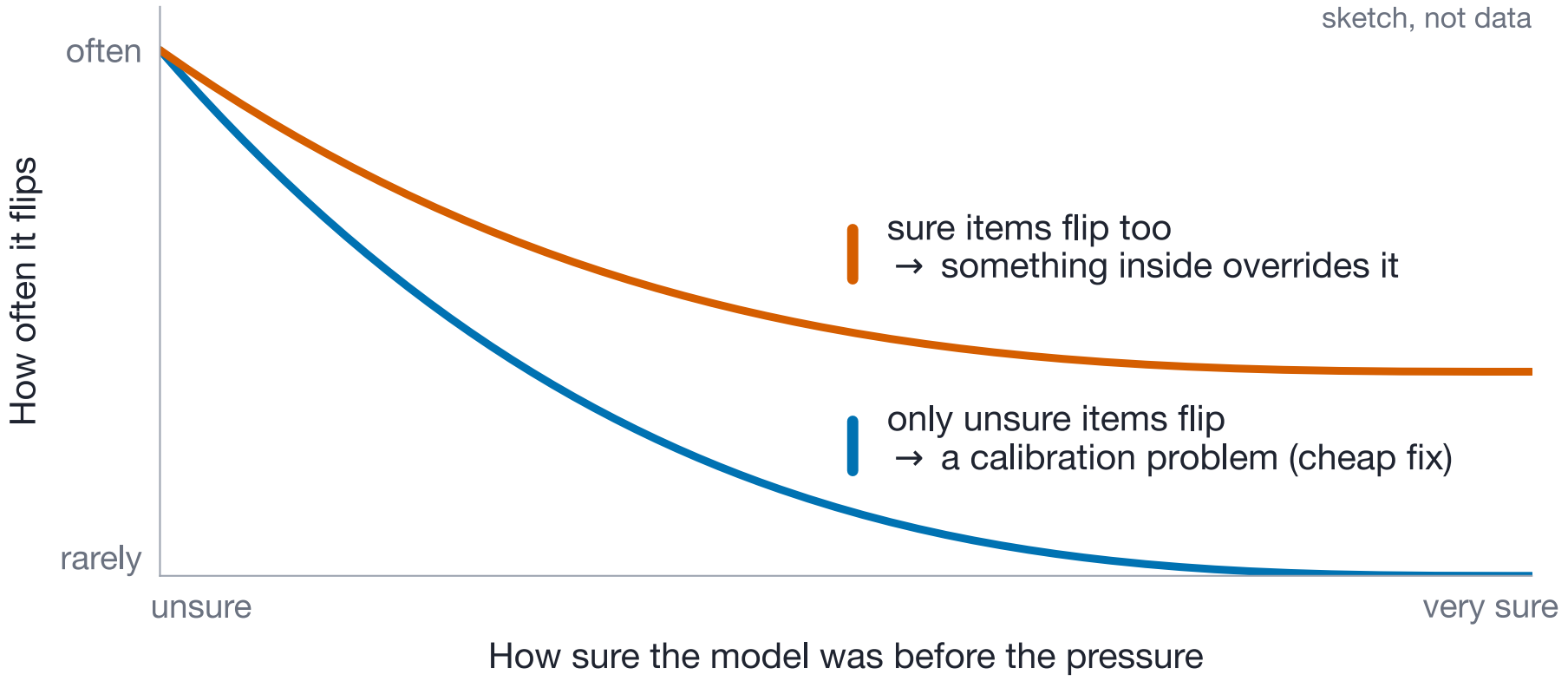
Two probes per item, as in KAPPA: one on the true answer (“knowing”), one on the answer the model gives (“acting”). Papers find 60–80° on average. If θ is the same on held and flipped items, we drop this idea.

Plan: sort first, fix last



Models: Gemma-2-9B first, then Qwen2.5-32B; 70B only if compute allows. Stage B starts from public checkpoint pairs (base vs instruct, OLMo SFT → DPO) because our own DPO is about 40–70 GPU-hours at 9B.

First run: a nearly free screen



→ Next, two quick checks before we trust any fate label: copy vs correctness, and planting fake fates (each under 2 GPU-hours).

One curve per pressure. It needs only the model's output scores, no internals.

Still open: which data?

Templated pushback

“I’m sure it’s Sydney” vs “I’m sure it’s Canberra”

- + same length, so patching lines up
- – artificial

TriviaQA questions

real questions the model mostly gets right

- + natural behavior
- – uneven lengths, harder to patch

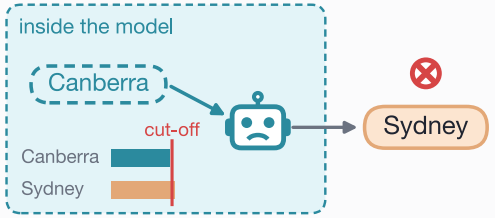
My draft: templates for the mechanism work, TriviaQA for behavior.

Questions for you

1. **Compute:** stay at 30B or below, or is 70B possible?
2. **Pressures:** start with a tempting wrong option + pushback?
3. **Speed:** how fast should we move, given a similar paper (2610.21708) and a crowded field?

Backup: six fates, each pointing to a different fix

a) Readout failure

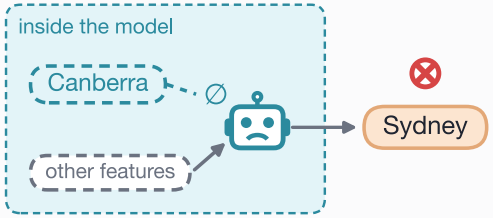


scores lean right; the cut-off is wrong

STUDIED logic task, ≤8B

why redo: untested on sycophancy or above 8B

b) Bypass

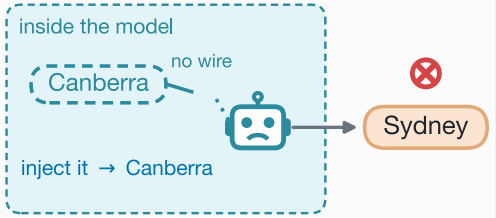


push it either way: nothing changes

STUDIED dates, agent asking, ≤14B

why redo: rarely for factual correctness

c) Not routed

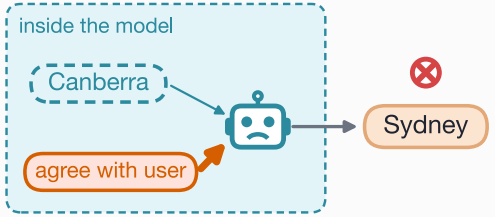


injected, it works; never carried there

STUDIED unanswerable questions, to 70B

why redo: natural use never measured

d) Override

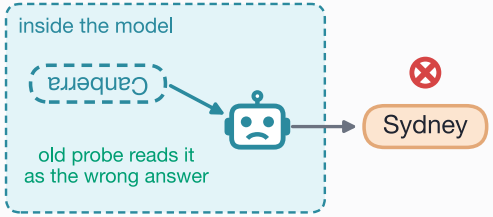


a competing signal wins

STUDIED sycophancy heads, ≤27B

why redo: unclear at 70B; not read as it agrees

e) Transformed

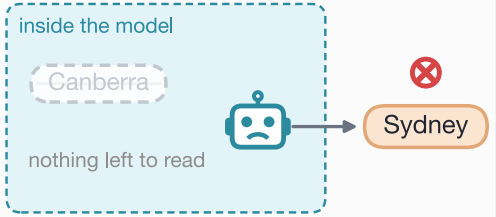


still there, re-encoded (sign-flipped)

BARELY once: instructed lying, Llama-2-70B

why redo: untested under user pressure

f) Erased



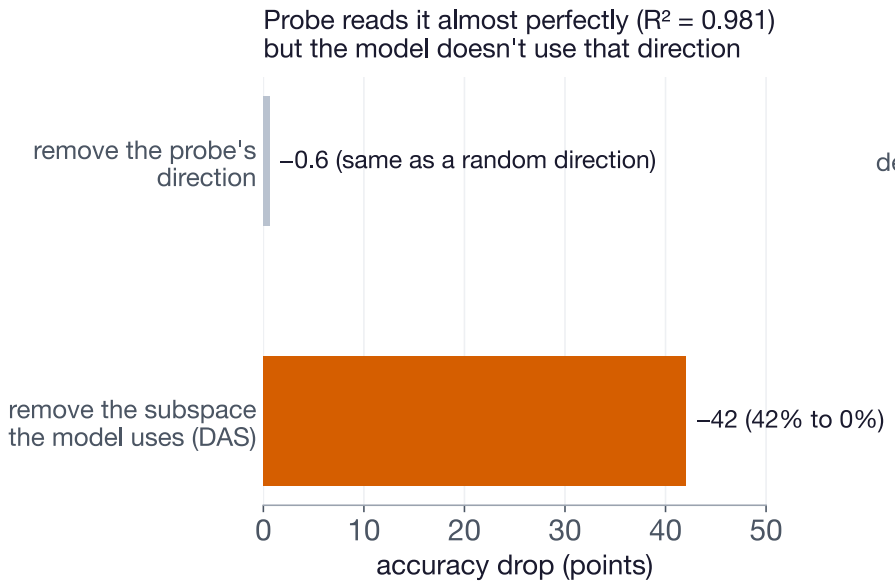
gone, even for a fresh probe

BARELY one contested claim, no controls

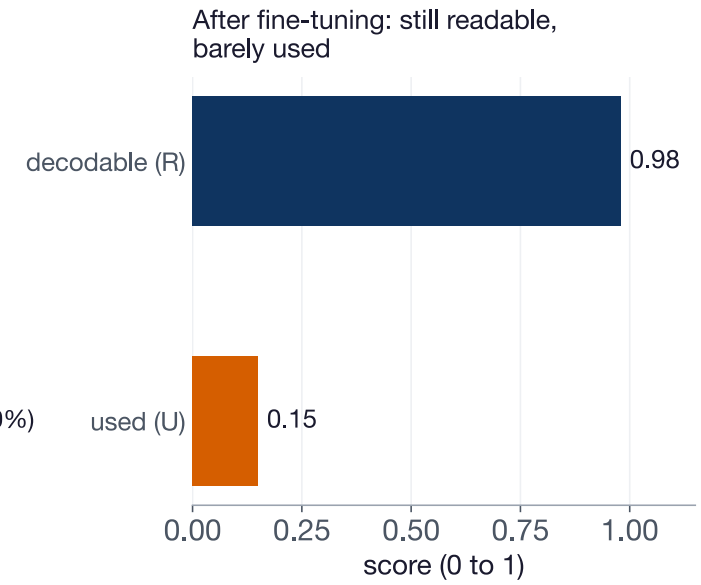
why redo: needs a controlled test

Dark band = studied before; dashed = barely studied. A fate label is only worth having if it predicts which fix works (prediction H2).

Backup: readable is not the same as used



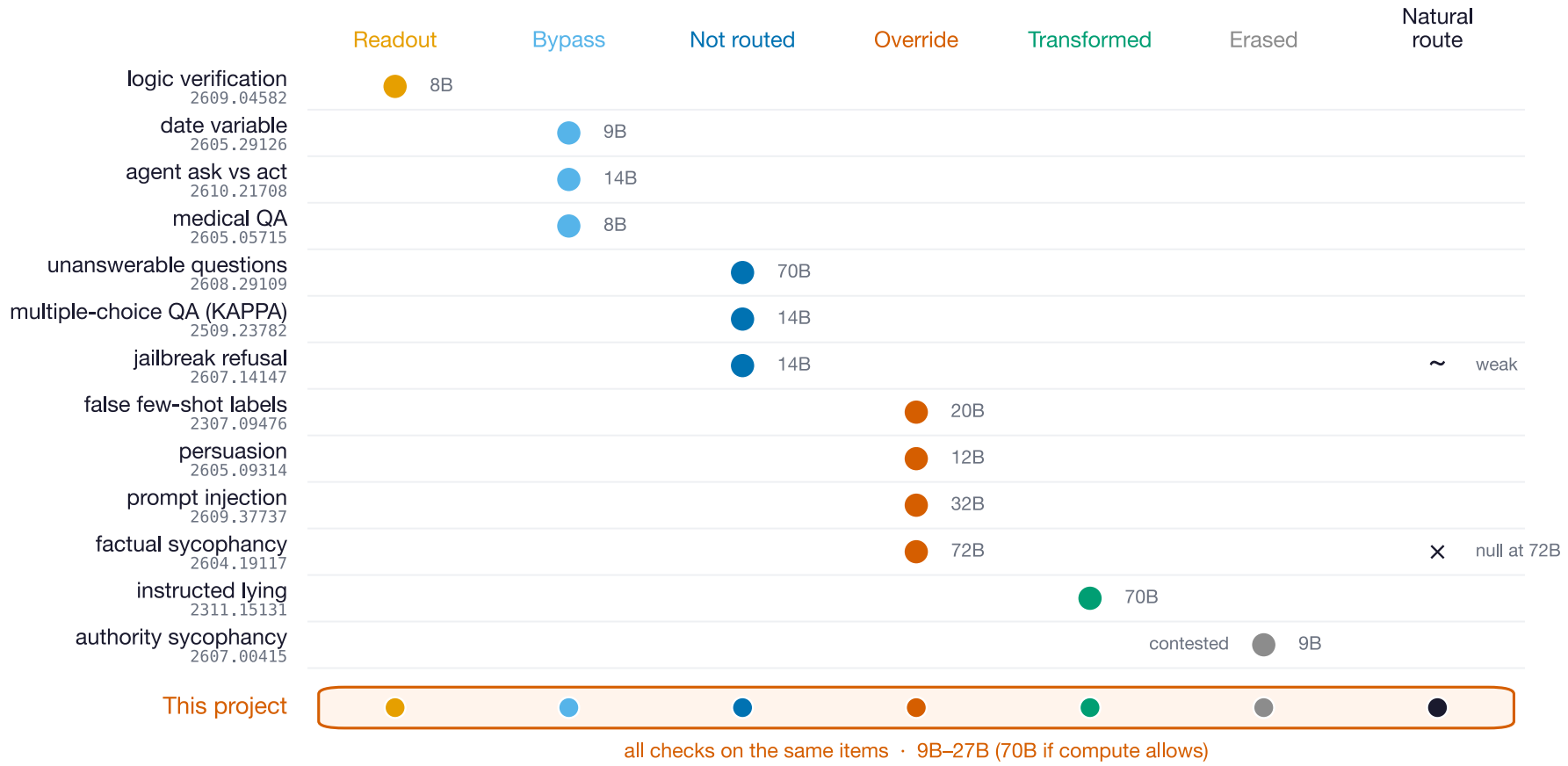
2605.29126 (Gemma-2-2B, a date variable)



2610.05292 (Qwen ≤4B): 'used' = does the decision change when only that fact changes?

→ So we never stop at "a probe can read it". Every claim gets a causal test: push the signal both ways, remove it, and swap inputs that differ only in that fact.

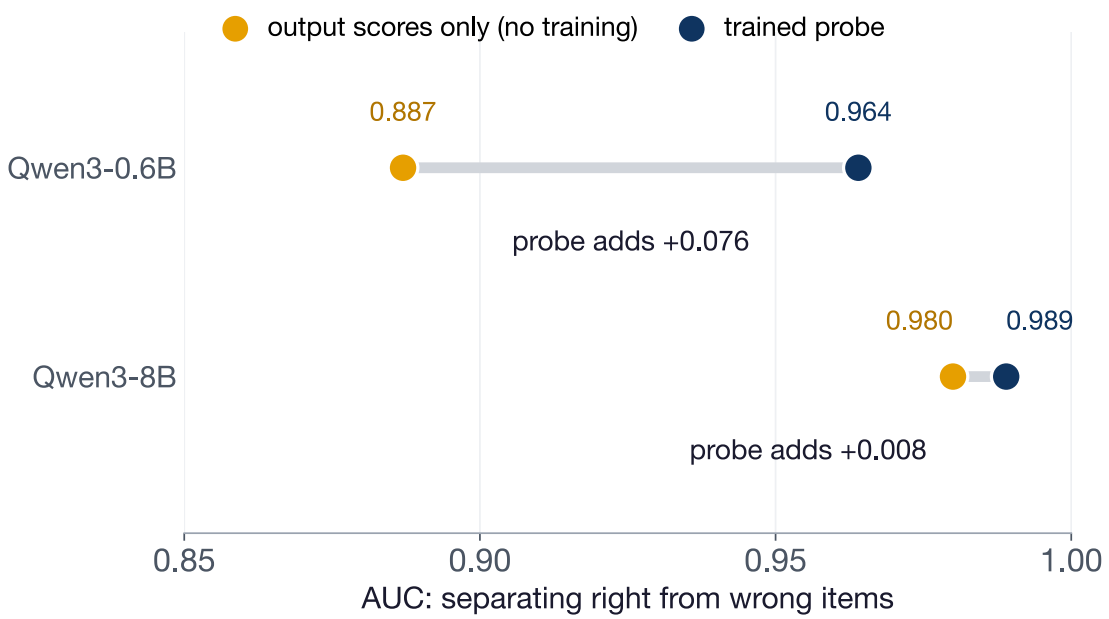
Backup: each paper tests one fate, with one method



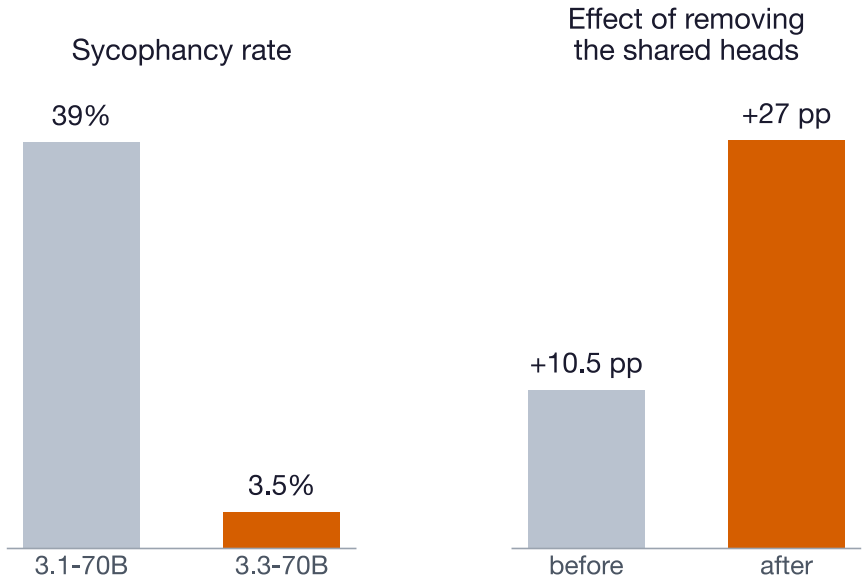
Three sycophancy papers disagree (“erased”, “override”, “two-stage”), each with a different method. Running every check on the same items tells us who is right.

Backup: two lessons from that work

1. Rule out the boring explanation first. At 8B, output scores alone almost match a trained probe.



2. Check whether a fix removed the problem. RLHF cut sycophancy ~10×, but the heads behind it matter more.



Llama-3.1-70B → Llama-3.3-70B (RLHF refresh), from 2604.19117

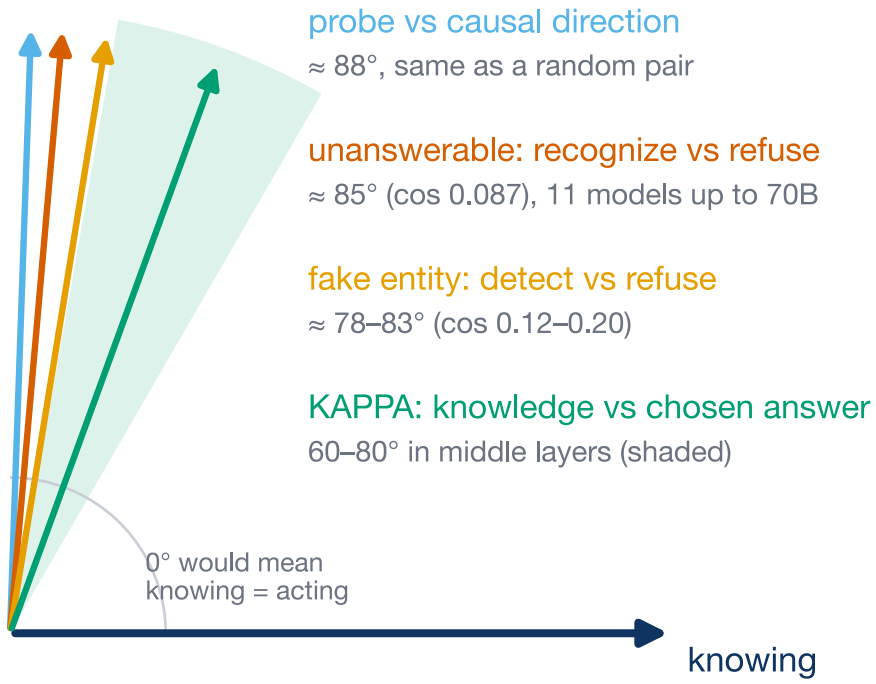
Left: 2609.04582 (synthetic logic task, up to 8B). Right: 2604.19117, one pair of public 70B checkpoints.

Backup: three predictions that can fail

	Prediction	If true	If false
H1 · Mechanism	Pushback and a tempting wrong option produce *different* fate mixes	“Sycophancy” is not one thing; each pressure needs its own fix	One shared mechanism; one fix may cover both
H2 · Fixes (tested last)	A fix matched to the fate beats plain DPO on held-out items, with less damage	Diagnose first, then fix	Skip the diagnosis; generic fixes are enough
H3 · Monitors	Probes trained on normal behavior miss “transformed” and “bypass” failures	Lie detectors need in-context checks	Today’s monitors are safer than feared

H1 is contested: 2601.11563 says one shared compliance direction, 2607.20146 finds separable sycophancy modes.

Backup: what papers measure for the angle



Nudging the “pick” direction toward the “true” one helps in KAPPA (TruthfulQA 56.7 → 67.8–73.5). Open: is the angle wider on the items where the model fails?

Backup: first experiments, ranked

#	Experiment	Question it answers	Cost
1	**Copy vs correctness**	After pushback, does the probe read the *fact*, or just the model's own earlier answer sitting in the chat?	1–2 h
2	**Plant fake fates**	Write known fakes (bypass, rotation, erasure…) into real activations. Which tools label them right?	< 1 h
3	**Output-score screen**	Flip rate vs prior confidence, per pressure. Do the output scores alone already explain the flips?	≈ 0 h

Kill rules: #1 fails Stage A's probe premise if AUC on the disagreeing cells is below 0.65; #2 bars any tool that miscalls more than 20% of plants. Cost = GPU-hours on Gemma-2-9B. Source: reports/Group geometry methods for fates.md, section 5.

Backup: the eight ranked ideas

#	Idea (geometry report, 25 Oct)	GPU-hours at 9B	Scoop
1	Copy vs correctness 2×2	1–2	low
2	Plant known fates, test the tools	< 1	none
3	Output-score screen + threshold push	≈ 0, then < 3	low
4	Do monitors survive? (frozen vs refit probes)	~2 + a DPO run	medium
5	Intervention ladder + rescue band by depth	5–10	high
6	One fix, or one per pressure?	< 3	medium
7	Override vs not-routed head classes	10–20	low–medium
8	Stage B: removed or only hidden?	40–70	moderate

Backup: why it’s worth the time

If it works, who uses it

- **Lie detectors:** shows *when* white-box monitors miss the failure (H3)
- **Audits:** if fixes only hide the problem, behavior tests can’t certify a model
- **Training:** flags fixes that teach hiding (weak detector + GRPO: >85% evasion; DPO: <25%)

Risks, and what we do about them

- **Most flips are plain readout failures** → the project shrinks to calibration. The free screen checks this first.
- **Scoop:** 2610.21708 (15 Oct) covers agent ask-vs-act → we stay on factual pressure, per-item fates and fixes.
- **No difference between fixes** → still a result: diagnosis isn’t needed.

Backup: how much to trust the literature review

- Fast pass: about 70 papers, many read from abstracts only; the search never met its stopping rule
- Found 22 of 31 papers we knew we had to find
- Headline numbers fact-checked; the rest still marked “to verify”
- No citation chasing yet, and several core papers unread, so we claim no novelty

Full report: reports/Knowledge action divergence internals round 2.md